CAMILLE

Athletic character sheet for Blender modeling and animation



LATEST DESIGN PASS

Adult fictional woman, 28. Muscular professional runner; defined shoulders, trunk and legs; firm lower abdomen; fuller glute transition; slightly reduced triceps; slender elbows and fingers; petite ears; blonde ponytail; fair neutral-warm complexion.

SCALE ANCHOR

5 ft 5 in / 165.1 cm, barefoot floor to anatomical skull vertex. The height is assigned by the user. Hair does not count toward stature.

HOW TO READ THE NUMBERS

Only stature and coordinate origin are fixed. Other numbers are working targets or visual estimates. Three decimals make Blender placement convenient; they do not mean the images support millimeter accuracy.

Use the front, side and rear as one visual family, then reconcile them against a single mesh. Separate image generations are not registered scans.

01 / Reference views

Front, anatomical left profile and rear are the current athletic design pass. Use the front and side for broad proportions, with the rear for back silhouette and glute form.







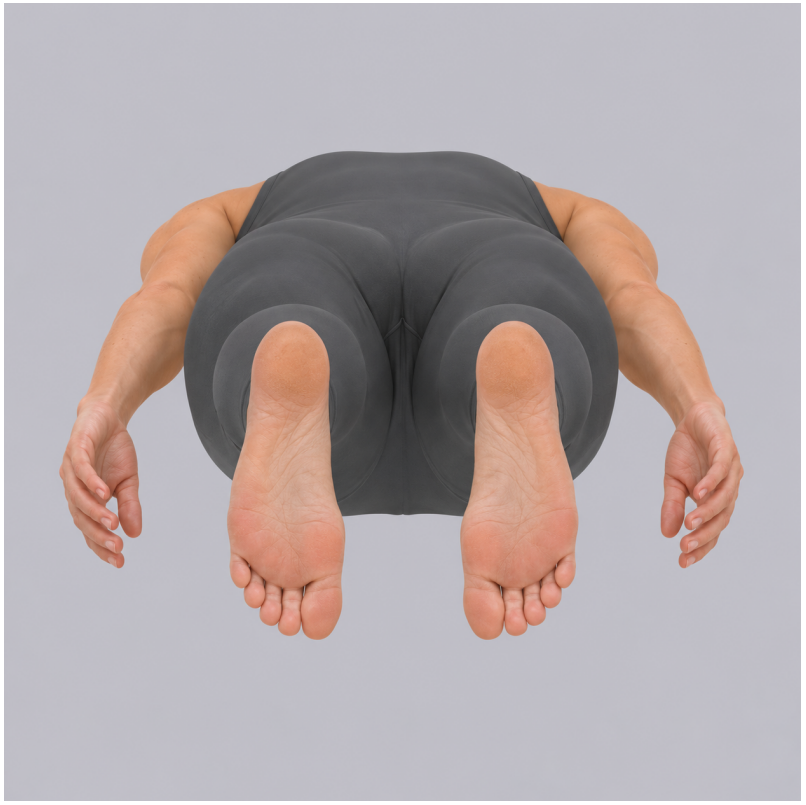
FRONT	LEFT PROFILE (NOSE IMAGE-LEFT)	REAR
Generated appearance reference; not calibrated orthographic.	Generated appearance reference; same-size registration is unverified.	Generated appearance reference; hidden depth is not measurable.

No circumference, exact skeletal placement, body-fat percentage or hidden anatomy can be extracted from these clothed renders. The user’s later local shape directions are represented as appearance goals, not verified centimeter changes.

02 / Best top and bottom references

Included at the user’s direction as the best views available. They provide appearance cues, especially hair silhouette, shoulder/trunk outline, feet and sole details.





TOP / OVERHEAD APPEARANCE	BOTTOM / UNDERSIDE APPEARANCE
Best generated top angle. Projection is inclined and not verified as 0-degree orthographic.	Best generated bottom angle. Perspective remains; not verified as 0-degree orthographic.

For final geometry, render both from the same completed mesh with orthographic cameras parallel to Z. Top camera looks down -Z; bottom looks up +Z. The floor should be hidden for bottom. These current images remain useful visual references, as approved.

03 / Blender scale and coordinate system

Set Blender units to Metric, Unit Scale 1.0. Use one Blender unit per meter. Place the model at real-world size; do not resize individual views to force apparent agreement.

SETTING	VALUE	PRACTICAL USE
World up	+Z	Vertical axis; floor at Z = 0.000 m
Character forward	-Y	Face, toes and sternum point toward negative Y
Anatomical left	+X	Camille’s left side; right side mirrors to negative X
Origin	(0, 0, 0)	Midpoint between foot contacts on floor plane
Stature	1.651 m	Floor to skull vertex; excludes hair
Hair envelope	about 1.657 m	Illustrative 6 mm allowance above skull; style-dependent
Skull vertex	Z = 1.651 m	Assigned stature endpoint
Chin underside	Z about 1.422 m	Working landmark; approximate
Anatomical head length	about 0.229 m	Skull vertex to chin; a design anchor
Head proportion	about 7.2 heads	165.1 / 22.9; anatomical head, not hair envelope

Front camera: look along +Y toward the face (camera located at -Y); profile camera from +X for anatomical left side; rear camera from +Y; top camera from +Z; underside camera from -Z. For reference-sheet registration, use orthographic cameras, identical target height and scale, and save camera transforms with each render.

Use the image landmarks only to block in. Once a single mesh is built, measure that mesh directly and replace provisional values with mesh-derived values.

04 / Whole-body landmark placements

Coordinates are starter targets for a neutral, relaxed pose. X values are centerline or bilateral offsets; Y is approximate depth and is least reliable from the pictures. Mirror bilateral points across X unless the final sculpt requires a documented asymmetry.

LANDMARK	X (m)	Y (m)	Z (m)	STATUS / USE
Floor / sole contact	+0.000	+0.000	0.000	Assigned origin
Ankle joint center	+0.055	+0.000	0.095	Estimated; verify against foot mesh
Knee joint center	+0.070	+0.000	0.445	Estimated patella / hinge center
Hip joint center	+0.085	+0.000	0.785	Estimated femoral head center
Crotch / perineal level	+0.000	+0.000	0.820	Approximate garment landmark; not a joint
Natural waist level	+0.000	+0.000	1.010	Approximate narrow waist level
Navel center	+0.000	-0.010	1.000	Visual estimate; Y is forward-negative
Lower sternum / xiphoid	+0.000	-0.015	1.190	Approximate centerline
Shoulder joint / glenohumeral	+0.205	+0.000	1.325	Estimated; mirror X for right
Elbow joint center	+0.160	+0.000	0.990	Estimated; arms relaxed
Wrist joint center	+0.135	+0.000	0.770	Estimated; arms relaxed
Middle fingertip	+0.145	-0.020	0.675	Estimated relaxed hand pose
Neck base / sternal notch	+0.000	-0.005	1.405	Approximate
Chin underside	+0.000	-0.030	1.422	Working design anchor
Eye pupil center	+0.030	-0.072	1.525	Face estimate; mirrored bilateral pair
Ear lobe	+0.073	+0.000	1.485	Provisional; current dainty ear not directly scaled
Ear superior helix	+0.073	+0.000	1.534	Provisional; verify on sculpt
Nose tip	+0.000	-0.105	1.505	Profile-derived direction only; Y uncertain
Anatomical skull vertex	+0.000	+0.000	1.651	User-assigned stature endpoint; excludes hair

Coordinate confidence: floor and assigned vertex are fixed by design. Joint centers and face depths are sculpting starts, not measured anatomy. Move joints to match the completed topology and intended range of motion before binding.

05 / Head and face targets

Use the ponytail head references for likeness and hairline cues. The current full-body pictures are too small to resolve eyelid, nostril, ear-fold or nail geometry. Face measurements are conditional estimates and should be fitted as ratios to the 165.1 cm stature.

FEATURE	WORKING TARGET	DEFINITION / CAUTION
Skull vertex to chin	22.9 cm	Working head-height anchor; crown excludes hair
Visible hair crown to chin	about 23.5 cm	Changes with ponytail/crown styling
Central hairline to chin	about 18.5 cm	Generated head-photo estimate; hairline is ambiguous
Pupil-center spacing	about 6.0 cm	Provisional face target; eye centers are mirrored
Apparent cheek-level skin width	about 12.7 cm	Surface width, not skeletal zygion-to-zygion breadth
Nose projection	1.3-2.0 cm; start 1.6 cm	Profile tip projection from subnasale, approximate
Ear height, current petite pass	about 4.7-5.0 cm	Provisional; latest smaller ear was not remeasured
Ear placement	upper rim near brow/pupil band; lobe above nose base	Use current body front/side; older close-ups show earlier larger ear
Mouth width	start at 4.5 cm	Art direction target only; fit to close-up
Nose base width	start at 3.2 cm	Art direction target only; fit to close-up
Eye opening width	start at 2.8 cm each	Art direction target only; sculpt eyelids around eyeball
Jaw / chin	soft tapered jaw; relaxed neutral chin	Shape cue, not numeric bone measurement

Face-to-mesh construction: establish eyeball centers first; shape orbital rims and lids around them. Place the ear canal/root in the side-head surface, then build helix, antihelix, concha, tragus and lobule. The user’s latest direction is smaller, lower, closer to skull; maintain natural attachment and avoid a protruding bowl silhouette.

Hairline, freckles and expression are surface/identity details. Ponytail mass is a separate hair object and must not change skull dimensions.

06 / Torso and athletic silhouette

The visual target is a trained runner: clear but natural muscle transitions, tight lower abdomen/adductors, athletic thighs and calves, and the latest user-requested glute mass. Keep musculature continuous and suitable for deformation; avoid sharp carved grooves that read as anatomy errors.

REGION	PLACEMENT / SHAPE TARGET	MEASUREMENT STATUS
Shoulder girdle	Acromion centers approximately X +/-0.205 m, Z 1.325 m	Estimated joint centers; refine on sculpt
Upper chest	Fullness follows current front/side silhouette; keep left-right even	Circumference unknown
Ribcage	Taper smoothly toward waist; preserve enough depth for breathing/rotation	Cross section not measurable from generated suit
Waist	Approximate centerline level Z 1.010 m	Height estimate only; circumference unknown
Pelvis	Hip joint centers near X +/-0.085 m, Z 0.785 m	Estimated centers; not outer hip landmarks
Glute / hamstring fold	Defined athletic transition; current rear/profile shows added mass	Shape cue only; mesh-derived depth later
Lower abdomen/adductors	Firm runner definition; smooth medial thigh separation	Aesthetic direction, not fat percentage or gap measurement
Body-fat change	User asked -2.5 percentage points from an unstated baseline	Absolute value cannot be assigned from these images

Do not reuse old bust, waist, hip or circumference values from earlier drafts. Those targets belonged to a taller, earlier character revision and were never measured. Record these only after the current mesh exists, using tape/curve circumference at named horizontal planes.

Suggested mesh measurement planes: fullest chest; directly below chest; narrowest waist; upper pelvis; maximum gluteal circumference; mid-upper arm; forearm maximum; proximal thigh; mid-thigh; calf maximum; ankle. Store plane height Z and circumference in the final JSON.

07 / Arms, hands, legs and feet

Use joint centers for the skeleton and let mesh surfaces define the visible muscle volume. The relaxed full-body references do not show a T-pose, so arm span and exact limb lengths should not be back-solved from screen pixels.

PART	STARTING TARGET / FORM	BLENDER MODELING NOTES
Upper arm	Shoulder Z 1.325 to elbow Z 0.990 m	Preserve slightly reduced triceps fullness; maintain deltoid/arm continuity
Forearm	Elbow Z 0.990 to wrist Z 0.770 m	Taper from elbow; slender but functional wrist
Elbow	Center about X +/-0.160, Z 0.990 m	User requests skinnier elbows; keep bony landmarks subtle
Hand	Wrist crease to middle fingertip 16.5-18.0 cm	Provisional; palms slender, fingers slightly longer/tapered
Fingers	Index, middle, ring, little: 3 phalanges each	Place joints at knuckles and flexion creases; middle longest; little shortest
Thumb	2 phalanges; saddle joint at palm	Opposition arc should meet each fingertip
Nails	Short natural oval/squoval plates	Surface details only; align to distal phalanx, subtle thickness
Hip / knee	Hip Z 0.785; knee Z 0.445 m	Starter centers; runner thigh volume is soft tissue around bones
Ankle / foot	Ankle Z 0.095 m; floor Z 0	Barefoot reference; toes point toward -Y
Foot length	Do not lock from image; placeholder range 23-25 cm	Perspective and stance make current pixels unsuitable for exact scale
Toes	Five digits, great toe largest; natural splay	Use toe bones only if close-up animation needs them

Finger proportions: position MCP knuckles as the metacarpal heads, then PIP and DIP joints for digits 2-5. Keep the middle finger longest; ring close behind; index shorter; little finger shortest. Use a three-bone chain only if required by the game rig, but preserve visible three-segment anatomy in the mesh. Thumb has CMC, MCP and IP articulation; CMC can be a control rather than a deform bone.

08 / Skeleton and joint-center starter map

These are rest-pose anchors, not a finished rig. Use Rigify as a starting generator, reposition its metarig joints to the mesh, generate the control rig, then validate skinning through motion tests. Keep deformation bones separate from animator controls.

BONE / CHAIN	LANDMARK	X m	Y m	Z m	ROLE / NOTE
root	ground / world	0.000	0.000	0.000	Non-deforming master
pelvis	pelvic center	0.000	0.000	0.790	Primary hip control
spine.01	lower abdomen	0.000	0.000	0.900	Deforms abdomen
spine.02	waist	0.000	0.000	1.010	Deforms waist
spine.03	lower chest	0.000	0.000	1.170	Rib cage
chest	upper chest	0.000	0.000	1.285	Upper torso / shoulder parent
neck	neck base to jaw	0.000	-0.005	1.405	Neck chain; add one mid-neck bone if needed
head	skull center to crown	0.000	0.000	1.535	Head pivot near atlanto-occipital joint, not at crown
clavicle.L	sternal end to shoulder	+0.105	0.000	1.365	Shoulder girdle
upper_arm.L	shoulder to elbow	+0.205	0.000	1.325	Mirror X for .R
forearm.L	elbow to wrist	+0.160	0.000	0.990	Mirror X for .R
hand.L	wrist to palm center	+0.135	-0.010	0.770	Palm axis points forward/down
thigh.L	hip to knee	+0.085	0.000	0.785	Mirror X for .R; preserve runner muscle
shin.L	knee to ankle	+0.070	0.000	0.445	Mirror X for .R
foot.L	ankle to ball	+0.055	-0.035	0.095	Forward is -Y
toe.L	ball to toe tip	+0.055	-0.165	0.035	Mirror X for .R

Bilateral chain: clavicle -> upper arm -> forearm -> hand; pelvis -> thigh -> shin -> foot -> toe. Mirror left/right positions across X. Add twist bones to upper arms and forearms for deformation if the game animation style benefits. Add heel/ball/toe controls only if footwear or running cycles require them.

09 / Blender setup and animation handoff

Recommended first pass: Blender’s official Rigify extension/add-on. It generates a human control rig from a repositioned metarig; it does not fit the body automatically. The official basic workflow is add metarig, position bones to the character and generate the rig. Bone structures distinguish deform bones from controls, so keep game export needs in mind.

STEP	ACTION	ACCEPTANCE CHECK
1. Scene scale	Metric units, 1 Blender unit = 1 m; floor at Z=0	Skull vertex exactly Z=1.651 m
2. Mesh blockout	One editable mesh aligned to front/side/rear; apply object transforms	Front and side proportions agree at shared landmarks
3. Metarig	Add Rigify human metarig; fit root, pelvis, spine, head, limbs, fingers	Joint centers sit inside correct anatomical pivots
4. Generate	Generate Rigify controls; save checkpoint before weight edits	No duplicate or broken generated rig
5. Skin weights	Bind mesh; inspect shoulder, hip, knee, elbow, wrist and fingers	No collapse, candy-wrapper twist, or suit tearing
6. Motion checks	Neutral pose, T/A pose, squat, stride, arm raise, wrist/finger curl	Silhouette remains continuous; feet plant properly
7. Game export	Decide target engine before final bone naming/export; bake actions as needed	Test a real imported animation in target engine

Plugin guidance: Rigify is enabled in local Blender 5.2.1 preferences. The Blender Extension “Unity Rig” is only relevant if the target is Unity Humanoid compatibility; it is not needed for modeling or a generic game rig. The Blender Extension Unity Rig is only needed for Unity Humanoid targets. MPFB was already installed and remains untouched; its generic base is not Camille.

Official references: Blender Manual - Rigify Basic Usage: https://docs.blender.org/manual/en/5.1/addons/rigging/rigify/basics.html | Blender Manual - Armature Structure: https://docs.blender.org/manual/en/5.0/animation/armatures/structure.html | Blender Extensions - Unity Rig: https://extensions.blender.org/add-ons/unity-rig/

10 / Measurements to replace after mesh fitting

The following are genuinely unknown from the current images. Use Blender measurements or a physical tape on the final mesh and record the chosen anatomical plane. Keep circumference loops horizontal and perpendicular to Z.

MEASURE	MESH VALUE	LANDMARK / METHOD
Bust circumference	_____ cm	Horizontal loop at fullest chest
Underbust circumference	_____ cm	Horizontal loop directly below chest
Natural waist circumference	_____ cm	Narrowest torso plane
Hip circumference	_____ cm	Maximum gluteal/hip plane
Neck circumference	_____ cm	Mid-neck horizontal loop
Upper arm circumference L/R	_____ / _____ cm	At maximum biceps/triceps girth
Forearm circumference L/R	_____ / _____ cm	At maximum forearm girth
Thigh circumference L/R	_____ / _____ cm	Record both proximal and mid-thigh planes
Calf circumference L/R	_____ / _____ cm	Maximum calf plane
Foot length L/R	_____ / _____ cm	Heel to longest toe in neutral stance
Arm span	_____ cm	Fingertip-to-fingertip with arms horizontal
Ear dimensions	H ____ / W ____ cm	Measure current mesh, both sides
Hand/finger lengths	_____ cm	Wrist crease to each fingertip; record each digit

After completing this page, replace the blank fields in the JSON and regenerate all front/side/rear/top/bottom views from the same mesh and saved orthographic cameras. This produces a true measurable character sheet rather than independent AI appearance images.

11 / Provenance, confidence and next build gate

This sheet is a production-oriented starting guide. It is not a medical anthropometric profile, scan, or certified rig specification.

LABEL	MEANING
ASSIGNED	User-selected fixed design anchor. Here: 165.1 cm stature and floor origin.
ESTIMATED	Landmark read or ratio inferred from generated references; camera and body scale remain approximate.
PROVISIONAL TARGET	A useful sculpting start chosen for this design; verify against the current mesh.
UNKNOWN	Cannot be determined from generated clothed views; measure the mesh after modeling.
APPROVED VISUAL	User accepted the best top/bottom attempts for visual guidance; their camera axes are not proven orthographic.

Quality gate before calling dimensions final

A single editable Camille mesh must exist. Measure its geometry in meters, name every measurement plane, align shared anatomical landmarks in all reference renders, save orthographic camera transforms, and re-render front/left-side/rear/top/bottom from that same mesh. Test the Rigify result with a walk/run cycle and hand poses. Replace every provisional value here with mesh-derived values, then archive the .blend and this updated PDF/JSON together.

Current exclusions: no absolute body-fat number; no verified circumference data; no exact 0-degree top/bottom camera proof; no cross-view pixel registration guarantee. The images are intentionally retained as visual references and should not be treated as a mesh measurement instrument.

Technical sources consulted

Blender Manual 5.1, Rigify Basic Usage - https://docs.blender.org/manual/en/5.1/addons/rigging/rigify/basics.html

Blender Manual 5.0, Armature Structure - https://docs.blender.org/manual/en/5.0/animation/armatures/structure.html

Blender Extensions, Unity Rig - https://extensions.blender.org/add-ons/unity-rig/ (optional only for Unity Humanoid targets)